

A Cap-Move Reformulation of Ling's Proof of Samuels' Conjecture

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Abstract

Let $0 \leq \mu_1 \leq \mu_2 \leq \dots \leq \mu_n$ and $\delta > 0$. Samuels' conjecture claims that if $X_1, \dots, X_n$ are independent non-negative random variables with $\mathbb{E}[X_i] = \mu_i$, then

$$\mathbb{P}\left(\sum_{i=1}^n X_i < \delta + \sum_{i=1}^n \mu_i\right) \geq \min_{1 \leq i \leq n} \prod_{j=i}^n \left(1 - \frac{\mu_j}{\delta + \sum_{k=i}^n \mu_k}\right).$$

This conjecture was recently proved by Ling [Lin26]. In this note, we provide an alternative presentation of Ling's proof via an operation called the *cap move*. This proof works directly with finitely supported distributions and avoids the reduction to the Bernoulli case.

1 Introduction

Let $X_1, \dots, X_n$ be independent non-negative random variables with $\mathbb{E}[X_i] = \mu_i$, with $0 \leq \mu_1 \leq \mu_2 \leq \dots \leq \mu_n$. Samuels' conjecture [Sam66, Sam68, Sam69] claims the following:

Conjecture 1 (Samuels' Conjecture). *For every $\delta > 0$,*

$$\mathbb{P}\left(\sum_{i=1}^n X_i < \delta + \sum_{i=1}^n \mu_i\right) \geq \min_{1 \leq i \leq n} \prod_{j=i}^n \left(1 - \frac{\mu_j}{\delta + \sum_{k=i}^n \mu_k}\right). \quad (1)$$

Samuels' conjecture implies several other conjectures such as Feige's [Fei04, Fei06], as well as a number of results in extremal combinatorics [AFH⁺12, FJ19].

Very recently Ling proved Samuels' conjecture [Lin26]. Ling's proof first reduces the problem to the Bernoulli case where each X_i takes the form of $w_i \text{Bern}(p_i)$, and then uses an induction argument by showing that a suitable coordinate can be removed one at a time. The current note provides a cap-move formulation of Ling's idea that works for general finitely supported distributions.

AI disclosure. This note benefits from discussions with GPT Sol 5.6 Pro, but is written entirely by the author.

2 Proof of Samuels' Conjecture

By approximating a general random variable with discrete ones, it suffices to consider the case where each X_i is a discrete random variable with finite support. In fact, a simple convexity argument can reduce the support size to 2, but the subsequent proof does not require this stronger result.

2.1 The Cap Move

A key ingredient of the proof is the following *cap move*: pick some $i \in [n]$ and $a \geq 0$, replace X_i by $X_i \wedge a := \min\{X_i, a\}$. Let p be the LHS probability of (1), and p_i^a be the counterpart after applying the cap move. Since a cap move makes the sum $\sum_{i=1}^n X_i$ no larger, we have $p_i^a \geq p$. The following lemma controls the increase from p to p_i^a .

Lemma 2.1 (Cap Move). *Let $S = \sum_{i=1}^n X_i$ with $\mathbb{E}[S] < \lambda$, and S_i^a be the counterpart after applying the cap move $X_i \mapsto X_i \wedge a$. Also define $p = \mathbb{P}(S < \lambda)$, $p_i^a = \mathbb{P}(S_i^a < \lambda)$, and*

$$b(S) := \mathbb{E}[S] - \mathbb{E}[S \mid S < \lambda], \quad (2)$$

$$r(S) := \lambda - \mathbb{E}[S \mid S < \lambda]. \quad (3)$$

Then $b(S) \geq b(S_i^a)$, and

$$\frac{p_i^a}{p} \leq \frac{r(S)}{r(S) - (b(S) - b(S_i^a))}.$$

Proof. Write $S_{-i} = \sum_{j \neq i} X_j$, which is independent of X_i . Then

$$\begin{aligned} \mathbb{E}[(\lambda - S_i^a)_+] - \mathbb{E}[(\lambda - S)_+] &= \mathbb{E}[\mathbf{1}_{\{X_i > a\}}[(\lambda - S_{-i} - a)_+ - (\lambda - S_{-i} - X_i)_+]] \\ &\leq \mathbb{E}[\mathbf{1}_{\{X_i > a\}}(X_i - a)\mathbf{1}_{\{\lambda - S_{-i} - a > 0\}}] \\ &= \mathbb{E}[\mathbf{1}_{\{X_i > a\}}(X_i - a)] \cdot \mathbb{P}(S_{-i} + a < \lambda) \\ &\leq \mathbb{E}[(X_i - a)_+] \cdot p_i^a. \end{aligned}$$

Since $\mathbb{E}[(\lambda - S)_+] = pr(S)$, $\mathbb{E}[(\lambda - S_i^a)_+] = p_i^a r(S_i^a)$, and

$$r(S) - r(S_i^a) = b(S) - b(S_i^a) - \mathbb{E}[(X_i - a)_+], \quad (4)$$

rearranging gives the target inequality. The claim follows from this inequality and $p_i^a \geq p$. $\square$

By Lemma 2.1, for a cap move that does not increase p by too much, we want a large $r(S)$ and a small $b(S) - b(S_i^a)$. This motivates the following notion of *safe move*.

Definition 2.2 (Safe Move). *We call a cap move safe if $r(S_i^a) \geq r(S)$.*

Before we proceed, we comment on the consequences of a safe move. First, by definition, the quantity $r(S)$ is non-decreasing under safe moves. Second, by (4), a safe move implies $b(S) - b(S_i^a) \leq \mathbb{E}[(X_i - a)_+] = \mathbb{E}[X_i] - \mathbb{E}[X_i \wedge a]$; therefore, after any finite sequence of safe moves, the sum of such terms with index i is at most $\mathbb{E}[X_i] = \mu_i$. Finally, over a finite sequence of safe moves, the sum of all b -decrements is at most $b(S)$. These properties turn out to be crucial in the final proof.

2.2 Existence of Safe Moves

The second step is a key claim in the proof: unless all X_i 's are deterministic, a non-trivial safe move always exists. This follows from the following lemma.

Lemma 2.3. *Assume the conditions and notations of Lemma 2.1. For each $i \in [n]$, there exists a nonnegative measure ν_i on $\mathbb{R}_+$ such that*

$$\text{Cov}(S, X_i \mid S < \lambda) = \int_0^\infty (r(S_i^a) - r(S))\nu_i(da).$$

In addition, if $\text{supp}(X_i) = \{x_1, \dots, x_m\}$ with $x_1 < \dots < x_m$, then $\text{supp}(\nu_i) = \{x_1, \dots, x_{m-1}\}$.

Proof. A simple proof uses an exponential tilt. For $\theta \geq 0$, let P_i be the distribution of X_i , and $P_{i,\theta}(dx_i) \propto e^{-\theta x_i} P_i(dx_i)$ be its exponential tilt. Also write $S_{i,\theta}$ be the counterpart of S when only P_i is replaced by $P_{i,\theta}$. By standard properties of exponential tilts,

$$\begin{aligned}\frac{d}{d\theta} r(S_{i,\theta}) \Big|_{\theta=0} &= \text{Cov}(S, X_i \mid S < \lambda), \\ \frac{d}{d\theta} P_{i,\theta}(dx) \Big|_{\theta=0} &= (m_i - x) P_i(dx),\end{aligned}$$

with $m_i := \mathbb{E}[X_i]$. To relate the LHS quantities, note that

$$\begin{aligned}r(S) &= \lambda - \mathbb{E}[S \mid S < \lambda] = \frac{\mathbb{E}[(\lambda - S)_+]}{\mathbb{E}[\mathbf{1}_{\{S < \lambda\}}]} \\ \implies \frac{d}{d\theta} r(S_{i,\theta}) \Big|_{\theta=0} &= \int \frac{d}{d\theta} P_{i,\theta}(dx_i) \Big|_{\theta=0} \cdot \frac{\mathbb{E}[(\lambda - S_{-i} - x_i)_+] - r(S) \mathbb{P}(S_{-i} + x_i < \lambda)}{\mathbb{P}(S < \lambda)}.\end{aligned}$$

We next show that there exists a measure $\nu_{i,0}$ such that

$$(m_i - x) P_i(dx) = \int_0^\infty \nu_{i,0}(da) (P_i^a - P_i)(dx), \quad (5)$$

where P_i^a is the distribution of $X_i \wedge a$. To find $\nu_{i,0}$, we integrate both sides on $[0, t]$ for $t \geq 0$:

$$\begin{aligned}\mathbb{E}[(X_i - m_i) \mathbf{1}_{\{X_i > t\}}] &= \int_{(t, \infty)} (x - m_i) P_i(dx) \\ &= \int_{[0, t]} (m_i - x) P_i(dx) \\ &= \int_0^\infty \nu_{i,0}(da) (P_i^a - P_i)([0, t]) \\ &= \int_0^\infty \nu_{i,0}(da) \mathbb{P}(X_i > t) \mathbf{1}_{\{a \leq t\}} = \mathbb{P}(X_i > t) \nu_{i,0}([0, t]).\end{aligned}$$

Therefore, we can construct $\nu_{i,0}(dt) = dw_i(t)$, with $w_i(t) := \mathbb{E}[X_i - m_i \mid X_i > t]$. Clearly $w_i(0^-) = 0$ and $t \mapsto w_i(t)$ is non-decreasing, so $\nu_{i,0}$ is a nonnegative Stieltjes measure. Using (5) and the above displays, we have

$$\text{Cov}(S, X_i \mid S < \lambda) = \int_0^\infty \nu_{i,0}(da) \frac{\mathbb{P}(S_i^a < \lambda)}{\mathbb{P}(S < \lambda)} (r(S_i^a) - r(S)),$$

and absorbing the (strictly positive) ratio into $\nu_{i,0}$ completes the proof of the identity. The support follows from $\text{supp}(\nu_i) = \text{supp}(\nu_{i,0}) = \{x_1, \dots, x_{m-1}\}$ and the jump structure of the Stieltjes measure $dw_i(t)$. $\square$

The following corollary is then immediate.

Corollary 2.4 (Existence of a safe move). *Let $X_1, \dots, X_n$ be discrete with finite support, and not all deterministic. Then there exist $i \in [n]$ and $a \in \text{supp}(X_i)$, $0 \leq a < \|X_i\|_\infty$ such that the cap move $X_i \mapsto X_i \wedge a$ is safe.*

Proof. Summing Lemma 2.3 over $i \in [n]$ gives

$$0 \leq \text{Var}(S \mid S < \lambda) = \sum_{i=1}^n \int_0^\infty (r(S_i^a) - r(S)) \nu_i(da).$$

As a result, there exist $i \in [n]$ and $a \in \text{supp}(\nu_i)$ with $r(S_i^a) \geq r(S)$, i.e. the cap move $X_i \mapsto X_i \wedge a$ is safe. By the second statement of Lemma 2.3, we have $a \in \text{supp}(X_i)$ and $0 \leq a < \|X_i\|_\infty$. $\square$

2.3 Variational representation of Samuels' lower bound

The final ingredient is a variational representation of Samuels' lower bound.

Lemma 2.5 (Variational representation). *For $0 \leq \mu_1 \leq \dots \leq \mu_n$ and $\delta > 0$, it holds that*

$$\min_{b \geq 0} \min_{\substack{z_i \in [0, \mu_i], i \in [n] \\ z_1 + \dots + z_n \leq b}} \sum_{i=1}^n \log \left(1 - \frac{z_i}{\delta + b} \right) = \min_{1 \leq i \leq n} \sum_{j=i}^n \log \left(1 - \frac{\mu_j}{\delta + \sum_{k=i}^n \mu_k} \right).$$

Proof. Let $M_k = \sum_{i=k}^n \mu_i$ for $k \in [n]$, and $M_{n+1} := 0$. For a fixed $b \in [M_{k+1}, M_k]$ with $k \geq 1$, since $z \mapsto \log(1 - \frac{z}{\delta + b})$ is decreasing and concave, an inner minimizer is $z_i = \mu_i$ for $i \geq k+1$, $z_i = 0$ for $i < k$, and $z_k = b - M_{k+1}$. Therefore, for $b \in [M_{k+1}, M_k]$,

$$G(b) := \min_{\substack{z_i \in [0, \mu_i], i \in [n] \\ z_1 + \dots + z_n \leq b}} \sum_{i=1}^n \log \left(1 - \frac{z_i}{\delta + b} \right) = \log \left(1 - \frac{b - M_{k+1}}{\delta + b} \right) + \sum_{i=k+1}^n \log \left(1 - \frac{\mu_i}{\delta + b} \right).$$

On the current interval,

$$G'(b) = \frac{1}{\delta + b} \left[-1 + \sum_{i=k+1}^n \frac{\mu_i}{\delta + b - \mu_i} \right],$$

and the term in the bracket is decreasing in b . Therefore, $G'(b)$ can change sign at most once, and only from positive to negative; this shows that G is quasi-concave and $G(b) \geq G(M_k) \wedge G(M_{k+1})$ on $[M_{k+1}, M_k]$. This implies

$$\min_{0 \leq b \leq M_1} G(b) = \min_{1 \leq i \leq n} G(M_i) = \min_{1 \leq i \leq n} \sum_{j=i}^n \log \left(1 - \frac{\mu_j}{\delta + \sum_{k=i}^n \mu_k} \right).$$

The case $b \geq M_1$ is obvious: the sum constraint is inactive, and $G(b)$ is increasing in $b \geq M_1$. $\square$

2.4 Completing the proof

We are ready to complete the proof of Conjecture 1. Start from $S_0 := S = \sum_{i=1}^n X_i$, and assume that all X_i 's are discrete with finite support. By Corollary 2.4, as long as X_i 's are not all deterministic, there exists a safe move $X_i \mapsto X_i \wedge a$, and repeating such operations gives a sequence $S_0, S_1, S_2, \dots$. By Corollary 2.4, each cap move will remove one support from X_i , so the finite support assumption implies that these are only finitely many safe moves, and the final sum S_m is deterministic. Set $\lambda = \delta + \sum_{i=1}^n \mu_i$ and let $p_t = \mathbb{P}(S_t < \lambda)$; since $\mathbb{E}[S_t]$ is non-increasing, we have $S_m = \mathbb{E}[S_m] \leq \sum_{i=1}^n \mu_i < \lambda$, so $p_m = 1$. Also, the definition of safe moves implies $r(S_0) \leq r(S_1) \leq \dots \leq r(S_m)$, so Lemma 2.1 gives

$$\frac{p_t}{p_{t-1}} \leq \frac{r(S_0)}{r(S_0) - (b(S_{t-1}) - b(S_t))}.$$

Let $z_i = \sum_{t \in [m]: i_t = i} (b(S_{t-1}) - b(S_t))$, with i_t being the coordinate capped at time t , the discussion below Definition 2.2 gives

$$0 \leq z_i \leq \mu_i, \quad \sum_{i=1}^n z_i \leq b(S_0).$$

Finally, a telescoping gives

$$\begin{aligned}
\log \mathbb{P}(S_0 < \lambda) &\geq \log \mathbb{P}(S_m < \lambda) + \sum_{i=1}^n \sum_{t \in [m]: i_t=i} \log \left(1 - \frac{b(S_{t-1}) - b(S_t)}{r(S_0)} \right) \\
&\geq \sum_{i=1}^n \log \left(1 - \frac{z_i}{r(S_0)} \right) = \sum_{i=1}^n \log \left(1 - \frac{z_i}{\delta + b(S_0)} \right) \\
&\geq \min_{b \geq 0} \min_{\substack{z_i \in [0, \mu_i], i \in [n] \\ z_1 + \dots + z_n \leq b}} \sum_{i=1}^n \log \left(1 - \frac{z_i}{\delta + b} \right).
\end{aligned}$$

The variational representation in Lemma 2.5 then proves Samuels' conjecture.

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